

CO1-AT3-TECHNICAL PRESENTATION

Name: T.Gobika

Reg.no:192312286

1.Learning Problems and Perspectives: Explain different types of learning problems in machine learning, including supervised, unsupervised, and reinforcement learning. Discuss various perspectives such as symbolic and statistical approaches. Highlight key issues like data quality, overfitting, and computational complexity, and explain how these factors influence the design and performance of learning systems.

Definition

Machine Learning is a branch of Artificial Intelligence (AI) that enables computers to learn from data and improve their performance without being explicitly programmed.

Types of Learning Problems

1. Supervised Learning

- Learns from labelled data (input with correct output).
- Used for classification and prediction.
- Example: Email spam detection.

2. Unsupervised Learning

- Learns from unlabelled data.
- Finds hidden patterns or groups in data.
- Example: Customer segmentation.

3. Reinforcement Learning

- Learns by interacting with the environment using rewards and penalties.
- Used in robotics and game playing.
- Example: Self-driving cars.

Learning Perspectives

➤ Symbolic Approach

Represents knowledge using rules and logic.

Easy to understand and explain.

➤ Statistical Approach

Uses probability and mathematical models to learn patterns from data.

Suitable for large and uncertain datasets.

Key Issues in Machine Learning

- Data Quality: Inaccurate or missing data reduces model performance.
- Overfitting: Model memorizes training data and performs poorly on new data.
- Computational Complexity: Large datasets require more processing time and memory.

Conclusion

The choice of learning type and approach depends on the problem. Good-quality data, avoiding overfitting, and efficient computation improve the performance of machine learning systems.

2. Concept Learning and Hypothesis Representation: Describe the concept learning task and its importance in machine learning. Explain how hypotheses are represented and evaluated. Discuss general-to-specific ordering and the role of training examples in refining hypotheses. Provide insights into how concept learning helps in classification problems and knowledge extraction.

Concept Learning and Hypothesis Representation

Definition

Concept learning is the process of learning a target concept from training examples. It is mainly used in classification tasks.

Importance of Concept Learning

- Helps classify new data correctly.
- Extracts useful knowledge from examples.
- Improves decision-making in machine learning.

Hypothesis Representation

- A hypothesis is a rule that predicts whether an example belongs to a target concept.
- The learning algorithm searches for the hypothesis that best fits the training data.

General-to-Specific Ordering

- Begins with the most general hypothesis.
- Gradually becomes more specific by using training examples.
- Removes incorrect hypotheses until the best one is found.

Role of Training Examples

- Positive examples help generalize the hypothesis.
- Negative examples help specialize the hypothesis.
- More accurate training data leads to better learning.

Applications

- Email spam classification.
- Medical disease diagnosis.
- Face recognition.
- Loan approval prediction.

Conclusion

Concept learning enables machines to build accurate classification models by refining hypotheses using training examples, making it an important technique for knowledge extraction and decision-making in machine learning.